Supplementary Materials for "A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology"

This supplementary materials includes proofs of technical results and additional numerical examples supporting conclusions from Section 4 the main manuscript.

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1 Proofs

1.1 Proof of Lemma 3.2

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k}))}, f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. A similar technique can be applied to the second equation to obtain the result. □

1.2 Proof of Theorem 3.4

Proof. First, Lemma 3.2 allows us to write

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F^{-1}_Y \circ U_{(j)}$ where $U_{(j)} \sim \mathrm{Beta}(j, n_1 - j + 1)$.

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_x^{F^{-1}_X \circ I^{-1}_{r_{1,j}, N_1 - r_{1,j}+1} \circ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_x^{F^{-1}_X \circ I^{-1}_{r_{1,j}, N_1 - r_{1,j}+1} \circ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x)} d\, I_{r_{1,j}, N_1 - r_{1,j}+1}(F_X(y)) \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ I_{j, n_1 - j+1} \circ I^{-1}_{k, n_2 - k+1} \circ I_{r_{2,k}, N_2 - r_{2,k}+1} \circ F_X(x) - I_{r_{1,j}, N_1 - r_{1,j}+1} \circ F_X(x) \right\}^+ f_{X_{2,(r_{2,k})}}(x)\, dx
\end{aligned}$$

$$= \int_{-\infty}^{+\infty} \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ I_{r_{2,k},N_2-r_{2,k}+1} \circ F_X(x) - I_{r_{1,j},N_1-r_{1,j}+1} \circ F_X(x) \right\}^+ \; d\, I_{r_{2,k},N_2-r_{2,k}+1}(F_X(x))$$
$$= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt.$$

Condition (8) states that

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt$$
$$= \int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\} \; dt.$$

Thus,

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ \; dt$$
$$= \int_0^1 I_{j,n_1-j+1}(x) \; dI_{k,n_2-k+1}(x) - \int_0^1 I_{r_{1,j},N_1-r_{1,j}+1}(x) \; dI_{r_{2,k},N_2-r_{2,k}+1}(x)$$
$$= \int_0^1 I_x(j, n_1-j+1) \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)} \; dx$$
$$- \int_0^1 I_x(r_{1,j}, N_1-r_{1,j}+1) \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(k, N_2-k+1)} \; dx$$
$$= \int_0^1 \sum_{p=j}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2-k+1)} \; dx$$
$$- \int_0^1 \sum_{p=r_{1,j}}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2-r_{2,k}+1)} \; dx$$
$$= \sum_{p=j}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(k+p, n_2+n_1-p-k+1)}{B(j, n_2-k+1)}$$
$$- \sum_{p=r_{1,j}}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(r_{2,k}+p, N_2+N_1-p-r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.$$

□

1.3 Proof of Proposition 3.5

Proof. Specification of $1_i(X_{i,j} \geq X_{i,(N_i-n_i+1)})$ implies that $r_{i,j} = N_i + 1 - j$. When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0,1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased [Askitis, 2021], or n_2 or N_1 are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned}
&I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\
&\implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\
&\implies I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t.
\end{aligned}$$

Technical condition (8) holds and the conclusion of Theorem 3.4 applies. □

1.4 Proof of Proposition 4.2

Proof. First, in combination with the Definition 4.1 and the assumption $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, we have

$$F_{Y_1}(x)^{\frac{n_1}{N_1}} \geq F_{Y_2}(x)^{\frac{n_2}{N_2}} \implies x \geq F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right).$$

By applying the Equation 7 from the Lemma 3.2 we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \mathrm{P}\left(F^{-1}_{Y_1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{+\infty} \int_{F^{-1}_{Y_1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x)
\end{aligned}$$

$$= \int_{-\infty}^{+\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_{2,(n_2)}}(x)$$
$$= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt.$$

Based on the Definition 4.1, we have

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\geq \int_0^1 F_{Y_2}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) &= \int_0^1 F_{Y_1}\left(F_{Y_2}^{-1}(t^{\frac{1}{n_2}})\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&\leq \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} - t^{\frac{N_1}{N_2}} dt \\
&= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}.
\end{aligned}$$

□

1.5 Proof of Theorem 4.3

Proof. Equation 7 from Lemma 3.2 yields

$$g_{n_1,n_2}(x) = F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right) = \Phi^{-1}\left(\Phi\left(\frac{x-\mu_2}{\sigma_2}\right)^{\frac{n_2 N_1}{N_2 n_1}}\right)\sigma_1 + \mu_1.$$

Let A and R be as defined, respectively, in (11) and (13). We have,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(g_{n_1,n_2}\left(Y_{2,(n_2)}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right)$$

$$
\begin{aligned}
&= \mathrm{P}\left(F_{Y_1}^{-1}\left(F_{Y_2}(Y_{2,(n_2)})^{\frac{n_2 N_1}{N_2 n_1}}\right) < Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_A \int_{F_{Y_1}^{-1}\left(F_{Y_2}(x)^{\frac{n_2 N_1}{N_2 n_1}}\right)}^{x} dF_{Y_1,(n_1)}(y) dF_{Y_2,(n_2)}(x) \\
&= \int_A F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) - \int_{A^c} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) \\
&= \int_{-\infty}^{\infty} F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}} dF_{Y_2,(n_2)}(x) + R.
\end{aligned}
$$

We continue with the left-hand side of the last equality,

$$
\begin{aligned}
&\int_{-\infty}^{\infty} \left[F_{Y_1}(x)^{n_1} - F_{Y_2}(x)^{\frac{n_2 N_1}{N_2}}\right] dF_{Y_2,(n_2)}(x) = \int_0^1 \left[F_{Y_1}\left(F_{Y_2}^{-1}(t^{1/n_2})\right)^{n_1} - t^{N_1/N_2}\right] dt \\
&\quad = \int_0^1 \left(\Phi\left(\frac{\Phi^{-1}(t^{1/n_2})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}\right)\right)^{n_1} dt \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = \int_{-\infty}^{+\infty} \Phi\left(\frac{\sigma_2 x + \mu_2 - \mu_1}{\sigma_1}\right)^{n_1} \left(n_2 \Phi(x)^{n_2 - 1}\phi(x)\right) dx \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = n_2 \int_{-\infty}^{+\infty} \mathbb{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2 - 1} - X \le 0 \end{array} \;\middle|\; X = x \right) \phi(x)\, dx \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = n_2\, \mathbb{P}\left(\begin{array}{l} Z_{1,1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1} X \le \frac{\mu_2 - \mu_1}{\sigma_1}, \\ Z_{2,1} - X \le 0, \ldots, Z_{2,n_2 - 1} - X \le 0 \end{array} \right) \; - \; \frac{N_2}{N_1 + N_2} \\
&\quad = n_2\, F_{\mathbf{W}}\left(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, \frac{\mu_2 - \mu_1}{\sigma_1}, 0, \ldots, 0)\right) \; - \; \frac{N_2}{N_1 + N_2}.
\end{aligned}
$$

The distribution function $F_{\mathbf{W}}$ corresponds to random variable $\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$ where $\mathbf{\Sigma}$ is as defined in (14) and (15). The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2 - \mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0.

The bound on the remainder (17) follows from the argument in Section 4.2. □

1.6 Proof of Lemma 4.4

Proof. Notice that the cumulative distribution function of a multivariate normal vector admits a sequential representation in terms of conditional univariate normal distribution functions. In particular, for $\mathbf{W} = (W_1, \ldots, W_n)^\top \sim \mathcal{N}(\mathbf{0}, \Sigma)$, there exist conditional CDFs $F_j(\cdot)$ such that

$$\mathbf{F}_{\mathbf{W}}(\mathbf{w}) = \prod_{j=1}^{n} F_j(w_1, \ldots, w_j), \tag{1}$$

where

$$F_1(w_1) := \mathrm{P}(W_1 \le w_1), \qquad F_j(w_1, \ldots, w_j) := \mathrm{P}\left(W_j \le w_j \mid W_1 = w_1, \ldots, W_{j-1} = w_{j-1}\right), \quad j \ge 2,$$

and each F_j is the CDF of a univariate normal distribution.

where $\mathbf{W} = (W_1, W_2, \ldots, W_n)$, $\mathbf{w} = (w_1, w_2, \ldots, w_n)$.

Then in general, for any $k \in \{2, \ldots, n_1 + n_2 - 1\}$, the conditional distribution of W_k given $W_1, \ldots, W_{k-1}$ is

$$W_k \mid W_1 = w_1, \ldots, W_{k-1} = w_{k-1} \sim \mathcal{N}\left(\mu_k, \sigma_k^2\right)$$

with

$$\mu_k = \Sigma_{k,1:(k-1)} \Sigma^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)}$$

$$\sigma_k^2 = \Sigma_{k,k} - \Sigma_{k,1:(k-1)} \Sigma^{-1}_{1:(k-1),1:(k-1)} \Sigma_{1:(k-1),k}$$

Here: $\Sigma_{k,1:(k-1)}$ is a $1 \times (k-1)$ row vector: the covariances between W_k and $W_1, \ldots, W_{k-1}$ $\Sigma_{1:(k-1),1:(k-1)}$ is a $(k-1) \times (k-1)$ covariance matrix, $\Sigma_{k,k}$ is the kth element on the diagonal of the covariance matrix Σ, and $\mathbf{w}_{1:(k-1)} = (w_1, \ldots, w_{k-1})^T$.

Then we will show the the parameters μ and σ^2 of the conditional univariate normal

distribution in the set of Equation (18) are finite.

When $k \leq n_1$, by applying the Equation (**??**)

$$
\begin{aligned}
\mu_k &= \Sigma_{k,1:(k-1)} \Sigma^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}(k-2)+1 & -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}(k-2)+1 \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_k^2 &= \Sigma_{k,k} - \Sigma_{k,1:(k-1)} \Sigma^{-1}_{1:(k-1),1:(k-1)} \Sigma_{1:(k-1),k} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}
\end{aligned}
$$

$$= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1}$$

$$= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}$$

Then

$$\mathrm{P}(W_k \le w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1})$$

$$= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{\frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right)$$

$$= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1}(1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}{\sqrt{1 + \frac{\sigma_2^2}{\sigma_1^2}(1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}} \right)$$

$$= \Phi\left(\frac{\frac{\frac{\mu_2-\mu_1}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{1 + \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right)$$

$$= \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1 + \frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right)$$

$$= c_k$$

$$\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}}\right) = \Phi(0) = 0.5$$

where $\Phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, for $k \le n_1$, $c_k < 1$.

When $k = n_1 + 1$, the only difference from the case $k \le n_1$ is that $\Sigma_{k,\,1:(k-1)}$ is different. We can then apply a similar method as in the case $k \le n_1$ to compute μ_k and σ_k.

$$\begin{aligned}
\mu_k &= \Sigma_{k,1:(k-1)}\Sigma^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}$$

$$\begin{aligned}
\sigma_k^2 &= \Sigma_{k,k} - \Sigma_{k,1:(k-1)}\Sigma^{-1}_{1:(k-1),1:(k-1)}\Sigma_{1:(k-1),k} \\
&= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}$$

Then

$$
\begin{aligned}
&\mathrm{P}(W_k \le w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}{\sigma_1^2}}}{\sqrt{2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}{\sigma_1^2}}}} \right) \\
&= \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}{\sigma_1^2}}}} \right) \\
&= \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) = \Phi\left(- \frac{\mu_2-\mu_1}{\sigma_2} \right)
$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, for $k = n_1 + 1$, $c_k < 1$.

When $k > n_1+1$, we first compute $\Sigma^{-1}_{1:(k-1),1:(k-1)}$. Based on the structure of Σ in Equation (14), the $\Sigma_{1:(k-1),1:(k-1)}$ can be decomposed into

$$
\Sigma_{1:(k-1),1:(k-1)} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \tag{2}
$$

defined with the following structure

$$D_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1 \times n_1} \quad D_{12}^T = D_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-n_1-1)\times n_1} \tag{3}$$

$$D_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(k-n_1-1)\times(k-n_1-1)}.$$

By applying the Equation (**??**) and Schur complement [Zhang, 2006],

$$\Sigma^{-1}_{1:(k-1),1:(k-1)} = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}, \tag{4}$$

$$E_{11} = (D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \cdot \frac{1}{k-n_1} \cdot \begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix} \right)$$

$$\cdot \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \Bigg)^{-1} =$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \cdots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \cdots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 \end{bmatrix}^{-1}$$

$$= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}_{n_1\times n_1}$$

$$E_{12} = -(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$

$$\times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 2 \end{bmatrix}^{-1}$$

$$
\begin{aligned}
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\begin{bmatrix}
\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\
-\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\
\vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\
-\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1
\end{bmatrix} \\
&\times \begin{bmatrix}
\frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
\frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1}
\end{bmatrix}
\frac{1}{k-n_1}
\begin{bmatrix}
k-n_1-1 & -1 & \cdots & -1 \\
-1 & \ddots & \ddots & \vdots \\
\vdots & \ddots & \ddots & -1 \\
-1 & \cdots & -1 & k-n_1-1
\end{bmatrix} \\
&= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\begin{bmatrix}
\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\
-\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\
\vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\
-\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1
\end{bmatrix} \\
&\times \begin{bmatrix}
\frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
\frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1}
\end{bmatrix} \\
&= \begin{bmatrix}
-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}_{n_1\times(k-n_1-1)}
\end{aligned}
$$

Since $\Sigma_{1:(k-1),1:(k-1)}$ is a symmetric matrix, then $E_{12} = E_{21}^T$. Thus,

$$E_{21} = -D_{22}^{-1}D_{21}(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1-1)\times n_1}$$

$$E_{22} = D_{22}^{-1} + D_{22}^{-1}D_{21}(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1}$$

$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$= \frac{1}{k-n_1} \begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \end{bmatrix}$$

$$= \frac{1}{k-n_1}\begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}$$

$$= \begin{bmatrix} 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1)\times(k-n_1)}$$

Then

$$\mu_k = \Sigma_{k,1:(k-1)}\Sigma^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(k-1)\times 1}$$

$$= \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right] \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1(k-n_1-1)}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$\sigma_k^2 = \Sigma_{k,k} - \Sigma_{k,1:(k-1)}\Sigma^{-1}_{1:(k-1),1:(k-1)}\Sigma_{1:(k-1),k}$$

$$= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{(k-1)\times 1}$$

$$= 2 - \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right] \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$= 2 - \Bigg(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Bigg)$$

$$= 2 - \Bigg(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1} + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1} + 1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1} + 1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}$$

$$
+\begin{bmatrix}1 & \dots & 1\end{bmatrix}
\begin{bmatrix}
-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}
\begin{bmatrix}\frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix}
$$

$$
+\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}
\begin{bmatrix}
-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
\vdots & \ddots & & \vdots \\
\vdots & & \ddots & \vdots \\
-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}
\begin{bmatrix}1 \\ \vdots \\ 1\end{bmatrix}
$$

$$
+\begin{bmatrix}1 & \dots & 1\end{bmatrix}
\begin{bmatrix}
1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\
\vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\
-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{bmatrix}
\begin{bmatrix}1 \\ \vdots \\ 1\end{bmatrix}\Bigg)
$$

$$
=2-\Bigg(\begin{bmatrix}\frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\end{bmatrix}
\begin{bmatrix}\frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix}
-\begin{bmatrix}\frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\end{bmatrix}
\begin{bmatrix}\frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1}\end{bmatrix}
$$

$$
-\begin{bmatrix}\frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\end{bmatrix}
\begin{bmatrix}1 \\ \vdots \\ 1\end{bmatrix}
+\begin{bmatrix}1-\frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & 1-\frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\end{bmatrix}
\begin{bmatrix}1 \\ \vdots \\ 1\end{bmatrix}\Bigg)
$$

$$
=2-\left(\frac{\frac{\sigma_2^2}{\sigma_1^2}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}-\frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{2(k-n_1-1)n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}+\frac{k-n_1-1}{k-n_1}+\frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{(k-n_1-1)^2n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}\right)
$$

$$
=2-\frac{k-n_1}{k-n_1+1}-\frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
$$

Then

$$
\begin{aligned}
&\mathrm{P}(W_k \le w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}} \right) \\
&= \Phi\left(- \frac{\frac{\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}} \right) \\
&= \Phi\left(- \frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1) + \sigma_1^2(k-n_1)^2}}} \right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(- \frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1) + \sigma_1^2(k-n_1)^2}}} \right) = \Phi(0) = 0.5
$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, for all k, $c_k < 1$. Moreover, we show the parameters μ and σ^2 of the conditional univariate normal distribution appearing in Equation (18) are finite.

Now

$$
\begin{aligned}
c = \max \Big\{ & \mathrm{P}\left(W_{n_1+n_2-1} \le w_{n_1+n_2-1} \,|\, W_1 = w_1, \ldots, W_{n_1+n_2-2} = w_{n_1+n_2-2}\right), \\
& \mathrm{P}\left(W_{n_1+n_2-2} \le w_{n_1+n_2-2} \,|\, W_1 = w_1, \ldots, W_{n_1+n_2-3} = w_{n_1+n_2-3}\right), \\
& \ldots, \mathrm{P}(W_2 \le w_2 \mid W_1 = w_1), \mathrm{P}(W_1 \le w_1) \Big\}, \\
= \max \Big\{ & c_2, c_3, \ldots, c_{n_1+n_2-1}, \mathrm{P}(W_1 \le w_1) \Big\}
\end{aligned} \tag{5}
$$

Since all terms in the set above are the cumulative distribution functions of univariate normal distribution and $\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)$, then c is strictly less than 1.

$$
\mathbf{F}_{\mathbf{W}}(\mathbf{w}) \le c^{n_1+n_2-1} \tag{6}
$$

where $\mathbf{F}_{\mathbf{W}}(\mathbf{w})$ denotes the cumulative distribution function of a mean-zero multivariate normal random vector with covariance matrix Σ, evaluated at $\mathbf{w} \in \mathbb{R}^{n_1+n_2-1}$. □

1.7 Proof of Theorem 4.5

Proof. Let $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\sigma}_1$, and $\hat{\sigma}_2$ be consistent estimators of the population parameters μ_1, μ_2, σ_1, and σ_2, respectively, as defined in Section 4.2. By replacing the population parameters in $\mathbf{w}$ and Σ with these estimators, we obtain the sample-level analogs $\hat{\mathbf{w}}$ and $\hat{\Sigma}$. Specifically, we define:

$$
\hat{\mathbf{w}} = \left(\frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, \ldots, \frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, 0, \ldots, 0 \right) \in \mathbb{R}^{n_1+n_2-1},
$$

where the first n_1 entries correspond to the standardized mean difference, and the remaining $n_2 - 1$ entries are zero, reflecting the structure introduced in Section 4.2.

The estimated covariance matrix $\hat{\Sigma}$ has the block form:

$$
\hat{\Sigma} = \begin{bmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{bmatrix},
$$

where each submatrix is obtained by replacing the population parameters in Σ with their respective sample estimates. These submatrices retain the same structural forms as defined in Section 4.2.

Based on the Equation (16), we can rewrite

$$\begin{aligned}&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ &= n_2(\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma)),\end{aligned}$$

where $\mathbf{F_W}(\mathbf{w})$ and $\Phi(\mathbf{w}, \Sigma)$ both denote the cumulative distribution function of a mean-zero multivariate normal random vector with covariance matrix Σ, evaluated at $\mathbf{w} \in \mathbb{R}^{n_1+n_2-1}$. For notational convenience, We use $\Phi(\mathbf{w}, \Sigma)$ throughout the proof.

Consider the term $\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma)$. We perform a Taylor expansion of this expression with respect to the parameters μ_1, μ_2, σ_1, and σ_2.

$$\begin{aligned}\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx &\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\ &+ \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2)\end{aligned}$$

Then for the first derivative term $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$:

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_1} \\ &= -\frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i}\end{aligned}$$

By applying Leibniz rule and without loss of generality, we compute the $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_1}$

$$
\begin{aligned}
\frac{\partial \Phi(\mathbf{w};\Sigma)}{\partial w_1} &= \frac{\partial}{\partial w_1}\int_{-\infty}^{w_1}\cdots\int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(z_1,\ldots,z_{n_1+n_2-1})dz_{n_1+n_2-1}\ldots dz_1 \\
&= \int_{-\infty}^{w_2}\cdots\int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(w_1,z_2,\ldots,z_{n_1+n_2-1})dz_{n_1+n_2-1}\ldots dz_2 \\
&= \int_{-\infty}^{w_2}\cdots\int_{-\infty}^{w_{n_1+n_2-1}} f_{W_1}(w_1)\cdot f_{\tilde{\mathbf{W}}}(\tilde{w})dz_{n_1+n_2-1}\ldots dz_2 \\
&= \phi(w_1)\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})
\end{aligned}
$$

where $\phi(w_1)$ is the probability density function of univariate normal distribution with $\mu = 0$ and $\sigma^2 = \frac{\sigma_2^2}{\sigma_1^2} + 1$, $\Phi(\tilde{\mathbf{w}},\tilde{\Sigma})$ is the conditional cumulative distribution function of the remaining components given $W_1 = w_1$. By applying Schur complement [Zhang, 2006],

$$
\tilde{\mathbf{w}} = (w_2,\ldots,w_{n_1+n_2-1}) - (\frac{\sigma_2^2}{\sigma_1^2},\ldots,\frac{\sigma_2^2}{\sigma_1^2},\frac{\sigma_2}{\sigma_1},\ldots,\frac{\sigma_2}{\sigma_1}) * (\frac{\sigma_2^2}{\sigma_1^2}+1)^{-1} * (w_1)
$$

$\tilde{\mathbf{w}}$ is a vector with the first $n_1 - 1$ elements is $\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$ and the last $n_2 - 1$ elements are $-\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2+\sigma_2^2}$.

$$
\tilde{\Sigma} = \Sigma_{-1,-1} - \Sigma_{-1,1}\Sigma_{1,1}^{-1}\Sigma_{1,-1}
$$

where $\Sigma_{-1,-1}$ denotes the submatrix of Σ obtained by removing the first row and the first column, $\Sigma_{-1,1}$ denotes the column vector obtained by removing the first element from the first column of Σ, $\Sigma_{1,-1}$ denotes the row vector obtained by removing the first element from the first row of Σ, and $\Sigma_{1,1}$ denotes the top-left scalar element of Σ.

Let denote the form of the matrix $\tilde{\Sigma}$:

$$
\tilde{\Sigma} = \begin{bmatrix} \tilde{\Sigma}_{11} & \tilde{\Sigma}_{12} \\ \tilde{\Sigma}_{21} & \tilde{\Sigma}_{22} \end{bmatrix}
$$

$$\tilde{\Sigma}_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{(n_1-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4} \end{bmatrix}_{(n_1-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_1-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{12}^T = \tilde{\Sigma}_{21} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

$$= \begin{bmatrix} 2-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 1-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \dots & 1-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \dots & 1-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 2-\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Therefore, $\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$ is finite. Then for $i = 1, \dots, n_1$, $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}$ is the same as $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_1}$. By applying Lemma 4.4, we get

$$\begin{aligned} \frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1} &= -\frac{1}{\sigma_1}\sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i} \\ &= -\frac{n_1}{\sigma_1}\cdot\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) \\ &\le O_P(n_1 c^{n_1+n_2-1}) \end{aligned}$$

By applying the similar method used for $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, we have

$$\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2} = \sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}\cdot\frac{\partial w_i}{\partial\mu_2}$$

$$
\begin{aligned}
&= \frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \\
&= \frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) \\
&= OP(n_1 c^{n_1+n_2-1})
\end{aligned}
$$

Notice that $\Phi(\mathbf{w}, \Sigma)$ has two parameters $\mathbf{w}$, and Σ involving σ_1, then using the Equation (137) in Petersen et al. [2008] and the similar method used for $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$, we have

$$
\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \sigma_1} + \operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1}) \\
&= -\frac{(\mu_2 - \mu_1) n_1}{\sigma_1^2} \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) + \operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})
\end{aligned}
$$

Before computing $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma}$, we define the $\phi(\mathbf{x})$ be the probability density function of $\Phi(\mathbf{w}, \Sigma)$, which is

$$
\phi(\mathbf{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} \exp\left[-\frac{1}{2}(\mathbf{x})^T \Sigma^{-1} (\mathbf{x})\right]
$$

then

$$
\log(\phi(\mathbf{x})) = -\frac{1}{2}(\log(\det(2\pi\Sigma)) + \mathbf{x}^t \Sigma^{-1} \mathbf{x})
$$

then using the Equation (57) and (61) in Petersen et al. [2008], we have

$$
\begin{aligned}
\frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} &= -\frac{1}{2}(\frac{\partial \log(\det(2\pi\Sigma))}{\partial \Sigma} + \frac{\partial \mathbf{x}^t \Sigma^{-1} \mathbf{x}}{\partial \Sigma}) \\
&= -\frac{1}{2}(\Sigma^{-1} - \Sigma^{-1} \mathbf{x}\mathbf{x}^t \Sigma^{-1})
\end{aligned}
$$

Then for the term $\operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})$ is:

$$
\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} &= \frac{\partial \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) d\mathbf{x}}{\partial \Sigma} \\
&= \int_{-\infty}^{\mathbf{w}} \frac{\partial \phi(\mathbf{x})}{\partial \Sigma} d\mathbf{x} \\
&= \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} d\mathbf{x} \\
&= -\frac{1}{2}\Sigma^{-1}\Phi(\mathbf{w}, \Sigma) + \frac{1}{2}\Sigma^{-1} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} \\
&= \frac{1}{2}\Sigma^{-1} \left(\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w}, \Sigma) \right)
\end{aligned}
$$

Then

$$
\begin{aligned}
\operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1}) &= \operatorname{tr}\left(\frac{1}{2}\Sigma^{-1} \left(\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w}, \Sigma) \right) \frac{\partial \Sigma}{\partial \sigma_1} \right) \\
&= \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_1} \right) - \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma) \frac{\partial \Sigma}{\partial \sigma_1} \right)
\end{aligned}
$$

To compute the $\operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})$, we firstly compute the $\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma) \frac{\partial \Sigma}{\partial \sigma_1} \right)$. Let define $\frac{\partial \Sigma}{\partial \sigma_1}$ be:

$$
\begin{aligned}
\frac{\partial \Sigma}{\partial \sigma_1} &= \begin{bmatrix} \frac{\partial \Sigma_{11}}{\partial \sigma_1} & \frac{\partial \Sigma_{12}}{\partial \sigma_1} \\ \frac{\partial \Sigma_{21}}{\partial \sigma_1} & \frac{\partial \Sigma_{22}}{\partial \sigma_1} \end{bmatrix} \\
&= \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}
\end{aligned}
$$

$$
A_{11} = \left[-\frac{2\sigma_2^2}{\sigma_1^3} \right]_{n_1 \times n_1}, A_{12}^T = A_{21} = \left[-\frac{\sigma_2}{\sigma_1^2} \right]_{(n_2-1) \times n_1}, A_{22} = \left[0 \right]_{(n_2-1) \times (n_2-1)}
$$

and by applying Schur complement [Zhang, 2006], we have

$$\Sigma^{-1} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}^{-1} = \begin{bmatrix} (\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & -(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \end{bmatrix}$$

where Σ_{11}, Σ_{12}, Σ_{21}, and Σ_{22} are defined in Equation (14).

By applying Equation (**??**), we have

$$\Sigma^{-1} = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$B_{11} = (\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - (1 - \frac{1}{n_2}) \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \dots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix} \right)^{-1}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{n_2\sigma_1^2}+1 & \frac{\sigma_2^2}{n_2\sigma_1^2} & \dots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \dots & \frac{\sigma_2^2}{n_2\sigma_1^2} & \frac{\sigma_2^2}{n_2\sigma_1^2}+1 \end{bmatrix}^{-1} = \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}_{n_1\times n_1}$$

$$B_{12} = -(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{n_1\times(n_2-1)}$$

$$B_{21} = -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$$

$$= -\begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2} + 1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2} + 1 \end{bmatrix}$$

$$= -\begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2} + 1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2} + 1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{(n_2-1)\times n_1}$$

$$B_{22} = \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$$

$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$
= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}
$$

$$
= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}
$$

$$
= \begin{bmatrix} 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}
$$

Then we have

$$
\Phi(\mathbf{w}, \Sigma) \leq c^{n_1+n_2-1}
$$

Therefore,

$$
\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_1} \leq \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}c^{n_1+n_2-1}
$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_1}$, we have

$$
\begin{aligned}
&\operatorname{tr}(\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}A_{11} + B_{12}A_{21}) + tr(B_{21}A_{12} + B_{22}A_{22}) \\
&= n_1\left(\frac{-\frac{2\sigma_2^2}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2}) + \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{\sigma_2}{\sigma_1^2}(n_2-1)\right) \\
&+ (n_2-1)\left(n_1\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{\sigma_2}{\sigma_1^2}\right) \\
&= -2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}
\end{aligned}
$$

Now we have

$$
\operatorname{tr}(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}) \leq 2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-1}
$$

Now we compute the term $\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right)$. Let rewrite the $\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}$ into matrix form:

$$
\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot x_1x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot x_nx_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot x_n^2 d\mathbf{x} \end{bmatrix}
$$

Define $\mathbf{w}_{-i}$ as the vector $\mathbf{w}$ with the ith component removed, and $\mathbf{w}_{-(i,j)}$ as the vector $\mathbf{w}$ with the ith and j th components removed. Similarly, $\mathbf{x}_{-i}$ and $\mathbf{x}_{-(i,j)}$ are defined in the

same manner.

Then

$$\begin{aligned}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i^2 d\mathbf{x} &= \int_{-\infty}^{w_i} \left(\int_{-\infty}^{\mathbf{w}_{-i}} \phi(\mathbf{x}_{-i}|x_i) d\mathbf{x}_{-i} \right) \; x_i^2 \; \phi(x_i) dx_i \\ &\leq c^{n_1+n_2-3} \cdot Var(X_i)\end{aligned}$$

$$\begin{aligned}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i x_j d\mathbf{x} &= \int_{-\infty}^{w_i} \int_{-\infty}^{w_j} \left(\int_{-\infty}^{\mathbf{w}_{-(i,j)}} \phi(\mathbf{x}_{-(i,j)}|x_i, x_j) d\mathbf{x}_{-(i,j)} \right) \; x_i x_j \; \phi(x_i, x_j) dx_i dx_j \\ &\leq c^{n_1+n_2-3} \cdot Cov(X_i, X_j)\end{aligned}$$

Replace $Var(X_i)$ and $Cov(X_i, X_j)$ using the elements of Σ, we have

$$\begin{aligned}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} &= \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix} \\ &\leq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\end{aligned}$$

where

$$S_{11} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) \end{bmatrix}_{n_1 \times n_1}$$

$$S_{12}^T = S_{21} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \end{bmatrix}_{(n_2-1)\times n_1}$$

$$S_{22} = \begin{bmatrix} 2c^{n_1+n_2-3} & c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} & 2c^{n_1+n_2-3} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Then,

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} = c^{n_1+n_2-3} * \Sigma$$

Therefore, followed by the results before,

$$\begin{aligned} & \operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ & \leq \operatorname{tr}\left(\Sigma^{-1}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ & = \operatorname{tr}\left(\Sigma^{-1}c^{n_1+n_2-3} * \Sigma * \Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \end{aligned}$$

$$= c^{n_1+n_2-3} \operatorname{tr}\left(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}\right)$$
$$= 2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-3}$$

Then

$$\operatorname{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_1}) = \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial \Sigma}{\partial \sigma_1}\right)$$
$$= O_p(c^{n_1+n_2-3})$$

By applying the similar method used for $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}$, we have

$$\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_1} = -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + \operatorname{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_1})$$
$$= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + O_p(c^{n_1+n_2-3})$$
$$= O_P(n_1 c^{n_1+n_2-3})$$

Similarly, we compute $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2}$,

$$\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2} = \operatorname{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_2})$$
$$= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial \Sigma}{\partial \sigma_2}\right)$$

To compute $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2}$, firstly we have

$$\frac{\partial \Sigma}{\partial \sigma_2} = \begin{bmatrix} \frac{\partial \Sigma_{11}}{\partial \sigma_1} & \frac{\partial \Sigma_{12}}{\partial \sigma_1} \\ \frac{\partial \Sigma_{21}}{\partial \sigma_1} & \frac{\partial \Sigma_{22}}{\partial \sigma_1} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

$$C_{11} = \left[\frac{2\sigma_2}{\sigma_1^2}\right]_{n_1 \times n_1}, C_{12}^T = C_{21} = \left[\frac{1}{\sigma_1}\right]_{(n_2-1)\times n_1}, C_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

Then by applying Lemma 4.4, we have

$$\Phi(\mathbf{w}, \Sigma) \leq c^{n_1+n_2-1}$$

Then

$$\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_2} \leq \Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2} c^{n_1+n_2-1}$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_2}$, we have

$$\begin{aligned}
&\operatorname{tr}(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_2}) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}C_{11} + B_{12}C_{21}) + tr(B_{21}C_{12} + B_{22}C_{22}) \\
&= n_1\left(\frac{\frac{2\sigma_2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2})\right) - \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{1}{\sigma_1}n_1(n_2-1)
\end{aligned}$$

$$
\begin{aligned}
&- (n_2 - 1) \left(n_1 \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} \frac{1}{\sigma_1} \right) \\
&= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}
\end{aligned}
$$

Now we have

$$
\operatorname{tr}(\Sigma^{-1}\Phi(\mathbf{w}, \Sigma)\frac{\partial \Sigma}{\partial \sigma_2}) \leq \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} c^{n_1+n_2-1}
$$

$$
\begin{aligned}
&\operatorname{tr}\left(\Sigma^{-1} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_2} \right) \leq \operatorname{tr}\left(\Sigma^{-1} \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_2} \right) \\
&= c^{n_1+n_2-3} \operatorname{tr}\left(\Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_2} \right) \\
&= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} c^{n_1+n_2-3}
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2} &= \operatorname{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_2}) \\
&= \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} \frac{\partial \Sigma}{\partial \sigma_2} \right) - \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} \Phi(\mathbf{w}, \Sigma) \frac{\partial \Sigma}{\partial \sigma_2} \right) \\
&= O_p(c^{n_1+n_2-3})
\end{aligned}
$$

Then we put $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$, $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}$, $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}$, and $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}$ together and based on the definition of root sample-size consistent estimator,

$$
|\hat{\mu}_1 - \mu_1| = O_p(n_1^{-1/2}),
$$

$$|\hat{\mu}_2 - \mu_2| = O_p(n_2^{-1/2}),$$

$$|\hat{\sigma}_1 - \sigma_1| = O_p(n_1^{-1/2}),$$

$$|\hat{\sigma}_2 - \sigma_2| = O_p(n_2^{-1/2}),$$

we have

$$\begin{aligned}\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx & \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\ & + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2) \\ & = O_P(n_1^{\frac{1}{2}} c^{n_1+n_2-3})\end{aligned}$$

Finally,

$$\begin{aligned}&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ &= O_P(n_1^{\frac{1}{2}} n_2 c^{n_1+n_2-3} + n_1 n_2^{\frac{1}{2}} c^{n_1+n_2-3})\end{aligned}$$

□

2 Simulations

2.1 Simulation Results for Theorem 4.3

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.3 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (2) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.3 using mvtnorm package [Genz and Bretz, 2009] in R.

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation Prob	Theoretical Prob
10	40	400	600	-1	1	1.0	3	0.4000	0.4000099
10	20	400	600	0	0	1.0	3	0.3970	0.3970305
10	40	400	300	-1	1	1.0	3	0.5715	0.5714174
10	20	400	600	0	1	1.5	2	0.3645	0.3647783
10	40	400	600	0	1	1.0	3	0.3995	0.3999245
20	40	400	600	0	1	1.0	3	0.3995	0.3999803
10	40	400	600	-1	0	1.5	3	0.4000	0.3995149
20	40	400	600	0	0	1.5	2	0.2970	0.2962431
20	40	400	600	0	0	1.0	3	0.3750	0.3740938
10	20	400	600	-1	1	1.0	3	0.3990	0.4000059

Table 1: Simulation Results for Normal Example.

References

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